

Synthesized Nano composites of nano silica and reduced graphene with urea for nitrogen fertilizer capsule production and their evaluation

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Abstract

Finding the replacement of urea for nitrogen fertilization, mesoporous nano silica (mNs) and reduced graphene oxide (rGO)-based urea nanocomposites were synthesized. These composites were characterized by UV-spectroscopy, FTIR, XRD, FE-SEM and HR-TEM to examine surface functional groups, morphology, and structural composition. XRD spectrum confirmation with SAED image of nano silica and graphene oxide nanocomposites confirms the polycrystalline and crystalline nature with 45–70 nm crystal size. The SEM revealed that the modified surface of mesoporous nano silica and reduced graphene oxide are well-distributed clusters and are composed of targeted nutrient. Synthesized composites were evaluated on seed germination, crop productivity and N use efficiency (NUE). The grain yields, NPK uptake and nitrogen use efficiency in rice and wheat was significantly higher with mNs and rGO-based composites applied to supply @ 60 N Kg ha⁻¹ as well as led to nutrients availability and microbial properties by increasing NPK content in soil over conventional use of urea. Application of mNs for 120 kg N ha⁻¹ showed maximum 64.44% increase in rice and 59% in wheat grain yields over conventional fertilization (120 kg ha⁻¹ through urea). Besides these, significant increment in grain yields, NPK content and uptake by crops was recorded Over control. Maximum 75.71 and 91.36% increase in NUE by rice and wheat crops, respectively by rice and wheat, respectively compared with conventional fertilization (120 kg ha⁻¹ through ordinary urea)

1. Background

Particle size ranges between 1 and 100 nanometers (nm) and are manufactured to possess specific properties that are quite different than their ordinary counterparts are known as engineered nanomaterials (ENMs) [3;5]. Research and development on engineered nanomaterials, in agricultural applications, probably facilitated and framed the next stage of development of genetically modified crops (GMCs), animal production inputs, biocides, and precision farming system [19]. Nanotechnology is a developing field of research involving the design, fabrication, characterization, and application of nanomaterials, or systems in the nanoscale due to the unique size-dependent physical and chemical properties [24 50]. Their extremely small size and high surface area-to-mass ratio allow quantum effects to be used to control their physical and chemical properties. One of the key area drawn attention is the synthesizing of different metal nanomaterials for agricultural use [4 ; 35].

Nanotechnology is one of the promising fields of research, which opens up a wide array of opportunities in the present decade and is expected to give major breakthroughs in variety of fields in future [46] and shows the potential to revolutionize the agricultural industry [15; 43]. Nano-fertilizers or nanoencapsulation of nutrients might have properties that are more effective to crops, released nutrients on-demand, controlled release of chemicals fertilizers that regulate plant growth and enhances targeted activity [48, 49 ; 50]. As a result, engineered nanomaterials (ENMs) based carriers for nutrients fertilizer had been extensively studied, in which chitosan, montmorillonite, zeolite, reduced graphene oxide (rGO), Mesoporous nano silica (mNs) are some important nanomaterials along with some polymer

encapsulation is most applied technique [1, 62]. Mesoporous nano silica is a form of silica and reduced graphene oxide (rGO) are recently developed materials in nanotechnology that had been intensively explored in nano research due to their unique properties, such as high surface areas, large pore volumes, pore size, easy penetrability with a narrow distribution and tunable particle diameters [11, 21 ; 29].

Silica is the most abundant in soil next to oxygen and comprises 31.0% of its weight, 3–17.0% in soil solution [27]. It is most commonly found in soils in the form of solution as silicic acid (H_4SiO_4) [8, 9 ; 10] and is translocated in the form of mono silicic acid through xylem in rice plants. Being a dominant component of soil minerals, silicon has many important functions in environment. Although silicon is not considered as an essential plant nutrient because most plants can be grown from seed to plants without its presence [36]. Silicon can also alleviate imbalances between zinc and phosphorus supply [43; 45]. Recently a number of workers have shown that silicon can decrease the toxic effects of Al and Graphene in hydroponics culture in several species [17]. Previous studies on nanomaterials have revealed their remarkable properties such as nontoxicity, biocompatibility, stable mesoporous structures, large surface areas, tunable pore sizes and volumes with great diversity in surface functionalization which have made them excellent candidates for nutrients delivery systems [11].

Graphene can be used in agriculture and various sectors of high-tech and food industry, e.g. (I) as a plant growth stimulator and a component of fertilizers [7]. (II) in nanoencapsulation and smart-release systems [7 ; 31]. Graphene is newest allotrope of carbon with superior structural and thermal properties. These properties make graphene modifiable for various applications in environmental remediation or energy storage [3 ; 6]. The scientific literature includes different types of graphene related materials, including graphite oxide, reduced graphene oxide and graphene, all of which have nearly identical structures. They share surface oxygen as functional group and sp^2 regions. [16] observed that graphene is not suitable for direct use in some applications, such as in drinkable water treatment. Therefore, reduction process of graphite oxide is a crucial step in obtaining reduced graphene oxide (rGO) which possesses more surface defects and wrinkles than graphene [5 ; 7]. Whereas, the application of graphene far infrared heating raised watermelon seedlings were observed and recorded, including temperature and humidity, seedling growth, etc. The results show that in the process of far infrared heating of graphene, no disinfection, pesticide spraying, and fertilization are carried out, and roots were strong and seedlings were flourishing, which greatly reduces manual operation and production cost. The time of seedling cultivation was shortened. Compared with other heating methods, the grafted seedlings of watermelon were put out of from nursery 10–15 days in advance, and the growth was good. The problem of out season freezing injury was solved and graphene far infrared heating in greenhouse can achieve the constant temperature effect of 5–15 °C when outdoor low temperature is below – 2 °C, so as to make crops grow smoothly in winter and eliminate the doubt that out of season vegetables can not be planted [7].

The agricultural sector will benefit greatly from nanotechnology based tools that can boost crop yields by increasing fertilizer nutrients availability in soil as well as nutrient absorption capacity [12; 32]. As a result of nano-fertilizers formulations, can produced unique materials which will controlled release of nutrients in balanced manner for crops during whole span of growth. Therefore, studies was undertaken to find out

the influence of synthesized urea nitrogen composites with mesoporous nano silica and reduced graphene oxide on seed germination, growth and yields of rice and wheat crops grown on normal alluvial Mollisol of Tarai region soil which will provide systematic information on use of nano materials as nano fertilizers in agricultural crops.

2. Materials And Methods

2.1 Synthesis of nano composites.

To synthesize urea nano composites, method outlined by [29 ; 55] was adopted in the laboratory. Mesoporous nano silica and reduced graphene oxide were purchased from M/S Nano Research Lab H 21 Gopalpur, Jamshedpur, Jharkhand, India. The urea was loaded on mesoporous silica and reduced graphene oxide by sonication and centrifugation. Dry powder of composites was obtained by dessicator drying after centrifugation [20].

2.2 Synthesis of mesoporous nano silica (mNs) composites with urea

The wet impregnation method was used to synthesized composites of mesoporous nano silica with urea hybrid composites [42]. A magnetic stirrer was used to mix 100 ml distilled water and nano silica in reaction vessel and urea was added in mixture according to ratios of matrix and urea (Fig. 1). 0.015 g TSC (Resin 3) was introduced in above reactive mixture. Thereafter, 0.012 ml formaldehyde (35.0%) was mixed by following respective order. The pH was maintained less than 6 by adding dilute sulphuric acid in the reaction mixture and these were stirred for 3 hours. These reaction mixtures were freezed in a refrigerator and dried at 11⁰C for 3 hours to find nano composite powders [29].

2.3 Synthesis of nano reduced graphene oxide (rGO) composites with Urea

Ultra sonicator was used to disperse rGO in deionized water, resulting in an 800 ppm rGO suspension. Urea was added in this suspension according to ratios of matrix and urea (Fig. 1). Suspension of rGO and urea was stirred for 1 hour at 80°C. Then, this suspension has been diluted and run through sonicator. A composite membrane was created by layering of urea on top of rGO. The composite was ovened for 6 hours at 60°C to enhance cross linking between rGO and urea molecules and then soaked in water for 24 hours to eliminate the adhering physical material. After dissolution in ethanol and centrifugation, precipitate was freezed and dried at 0⁰C to find the composite powder.

2.4 Characterization of nano composites.

Characterization of synthesized nanocomposites i.e.X-Ray Diffraction (XRD) (BRUCKER D8 ADVANCE),, Field Emission Scanning Electron Microscopy (FE SEM/EDX) (CARL ZEISS SUPRA 55 operating at 10 kv), High Resolution Scanning Electron Microscopy (HR TEM) (JEOL JEM 2100 PLUS) were performed at IIT, Bombay with the help of Spring testing solutions. Fourier Transform Infrared Spectroscopy (FTIR)

(PERKIN ELMER FTIR C94012). Laboratory study on engrafting of urea on mNs and rGO has been performed in the Department of Physics (Nanotechnology Unit), College of Basic Science and Humanities (CBSH), G. B. Pant University of Agriculture & Technology, Pantnagar-263145, Uttarakhand, India. Characterizations of all prepared nanocomposites were done at IIT Bombay, Maharashtra through Spring Testing Solutions.

2.4 Estimation of urea nitrogen loading efficiency of nanomaterials

2.4.1 Loading efficacy of nano composites

For assessing loading efficiency of urea nitrogen, amount of matrix (mNs) or reduced grapheme (rGO) urea were added to prepare nanocomposites with predefined ratios (w/w) viz. Ratio 1:0.1 (1), 1 : 0.2 (2), 1:0 .5 (3), 1:0.6 (4), 1:0.7 (5), 1:0.8 (6), 1:0.9 (7), 1 : 1 (8), 1 : 1.1 (9), 1 : 1.2 (10), 1 : 1.4 (11) and 1 : 1.8 (12). Each composite sample was analyzed for its nitrogen content by micro Kjeldhals method [26]. The outcomes are explained in graphical format in Fig. 1.

2.4.2 Dilution effect on nitrogen content of nano composites

To asses dilution effect on synthesized nanocomposite fertilizers (mNs-Urea and rGO-Urea), various dilutions (1:10,1:15,1:20,1:40,1:50) of different nano composites : distilled water were made and each dilution mixture analyzed for nitrogen content by micro Kjeldhals method [26]. The results of nanocomposite fertilizers dilutions (1:10, 1:15, 1:20, 1:40, 1:50) are presented in Table 2.

2.5 Instruments used for characterization of nano composites.

Characterization of synthesized nanocomposites i.e.X-Ray Diffraction (XRD) (BRUCKER D8 ADVANCE),, Field Emission Scanning Electron Microscopy (FE SEM/EDX) (CARL ZEISS SUPRA 55 operating at 10 kv), High Resolution Scanning Electron Microscopy (HR TEM) (JEOL JEM 2100 PLUS) were performed at IIT, Bombay with the help of Spring testing solutions. Fourier Transform Infrared Spectroscopy (FTIR) (PERKIN ELMER FTIR C94012). Laboratory study on engrafting of urea on mNs and rGO has been performed in the Department of Physics (Nanotechnology Unit), College of Basic Science and Humanities (CBSH), G. B. Pant University of Agriculture & Technology, Pantnagar-263145, Uttarakhand, India. Characterizations of all prepared nanocomposites were done at IIT Bombay, Maharashtra through Spring Testing Solutions.

2.6 Seed germination analysis

Various treatment solutions were prepared from 1000 ppm solutions of both nano composites according to 14 treatment combinations that are described in para no. 2.7.3. For each treatment, amount of composites nano was calculated in ppm on the basis of their content and applied quantity (kg ha^{-1}) of

nitrogen in each treatment. Ten seeds of wheat (*triticum aestivum* L) variety PBW-2968 and rice (*Oriza sativa* L) variety PR-112) were soaked in each treatment solutions using a moist paper towel (22 × 23 mm) in triplicate. Seeds were placed on a single sheet of moist paper towel and covered with another single sheet of same paper towel. After that, this sandwich sheet was rolled, placed thereafter in a plastic beaker, and then covered with a polythene bag. The beaker was then placed in an incubator at $25 \pm 2^{\circ}\text{C}$ for 16 hours with light conditions. The total numbers of germinated seeds were counted day after day and germination observations were recorded up to 7th day in wheat and 5th day in rice in respective seasons.

2.6.1 Seed germination percentage

Germination testing was presented according to the guidelines issued by the International Seed Testing Association [25; 41].

$$\text{Germination Percentage} = \frac{N_i}{N_t} \times 100$$

Where- N_i is the total number of germinated seeds for a continuous period of seven days

N_t is the total number of seeds.

2.6.2 Seedling vigour index

Seedling vigor index has been calculated by the formula given by [8]. It was calculated as:

$$\text{Seedling vigor index} = \text{germination (\%)} \times \text{Seedling length (cm / seedling)}$$

2.7 Green house experiment

2.7.1 Description of soils : The soils of *Tarai* have developed from the alluvium brought down from mountains by numerous streams flowing through *Bhabhar* and *Tarai* regions. The soils of *Tarai* region are silty and loamy, having good moisture storage capacity and are highly productive. The soils for pot experiment in present study belongs to normal alluvial soil from CRC, Pantnagar (U.S Nagar) from 0–15 were analyzed in laboratory for their physico-chemical property. Results of initial soils analysis are given in following Table 1.

Table 1
Initial characteristics of experimental normal alluvial soil.

S.No	Property	Normal alluvial soil
1.	Soil texture class	Silty clay loam
2.	Bulk density(g cm ⁻³)	1.31
3.	Particle density(g cm ⁻³)	2.64
4.	Porosity (%)	50.37
5.	pH (1:2.5 soil: water suspension)	7.76
6.	EC (dSm ⁻¹)	0.28
7.	Organic carbon (%)	0.79
8.	Available N (kg ha ⁻¹)	263.70
9.	Available P (kg ha ⁻¹)	16.31
10.	Available K (kg ha ⁻¹)	126.40
11.	Total N %	0.127
12.	Moisture content (%)	14.23

2.7.2 Weather condition: Pantnagar fall under the sub- humid and subtropical climate zone with hot dry summers and cool winters. Generally, south west monsoon sets in the third or fourth week of June and continues up to the end of September with its peak in July. The mean annual rainfall is about 1400 mm, of which 80–90 per cent is received between June and September months. Frost generally occurs towards the end of December and may continue till the end of January. The daily average minimum temperature varies from 1.0 to 9.0 °C and maximum temperature varies from 30 to 43 °C.

2.7.3 Treatments and other details of pot experiment : During 2020-21 kharif and Rabi season, a pot experiment was conducted to evaluate urea-mNs and rGO composites in green house with 14 treatments namely ; T₁- N0P60K40, T₂- N0P40K40 + Maximum silica, T₃- N0P60K40 + Maximum Graphene, T₄-N30 (urea)-P60-K40, T₅-N60(urea)-P60-K40, T₆-N90(urea)-P60-K40, T₇-N120(urea)-P60-K40, T₈-N30(Urea-mNS)-P60-K40, T₉-N60(Urea-mNS)-P60-K40, T₁₀-N90(Urea-mNS)-P60-K40, T₁₁-N120(Urea-mNS)-P60-K40, T₁₂-N30(urea-rGO)-P60-K40, T₁₃-N60(urea-rGO)-P60-K40, T₁₄-N90(urea-rGO)-P60-K40 and T₁₅-N120(urea-rGO)-P60-K40 in completely randomized block design (CRD) with three replication for both crops. Three rice (*Oriza sativa* L) seedlings variety-PR113 were transplanted on 27.06.2020 after pulverizing soil and placing fabricated nano nitrogen capsules in root zone and crop was harvested on 16.10.2020 after maturity. In same pots, four seeds of wheat (*Triticum aestivum* L.) variety PBW-2967 were sown on 21.11.2020 after tilling soil fabricated nano nitrogen capsules placed in soil. In ordinary

urea treatments, half nitrogen as basal at the time of sowing and half nitrogen were applied in two splits. Plant and soil samples were collected after harvesting of rice and wheat crops.

2.7.4 Statistical Analysis: The data collected during the experimental period and after the completion of experiment were statistically analyzed with the help of the following statistical techniques.

3. Results And Discussion

3.1 Characterization and verification of urea nitrogen composites with nano silica and reduced graphene

Urea-mNs and urea-rGO powder was obtained after purification and drying of these composites. These composites were characterized by using Fourier Transform Infrared Spectroscopy (FTIR), X-ray diffractometer (XRD), Scanning electron microscopy (SEM), Transmission electron microscopy (TEM) and Energy-dispersive X-ray (EDX) spectroscopy, analysis.

3.1.1 FTIR analysis

FTIR analysis is carried out using PERKIN ELMER FTIR C94012. The FTIR range of mNs-Urea and rGO-Urea is outlined in Fig. 1a, appears a wide peak showed up at 3348 cm^{-1} and 3334 cm^{-1} (Fig. 2a and b, respectively) in the tall recurrence region credited to the nearness of nitrogen (urea) particle because it is an apparent crest of N-H stretching vibrations. The C = O extending recurrence is showed up at around 1635 cm^{-1} in both the cases. The wide peak from 3000 to 3500 cm^{-1} (Fig. 1a and b) also relegated to the nearness of O-H bond. Additionally, a peak comparing to vibration bowing can be taken note at 1061 cm^{-1} (Fig. 1a), which demonstrates the the nearness of siloxane extending bonds of mesoporous nano silica. The sharp peak found at 1635 cm^{-1} (Fig. 1b) could be a reverberation peak that can be doled out to extending and stretching vibration of bonds of water atoms adsorbed on reduced graphene oxide. This demonstrates that the urea has facilitated with mesoporous nano silica and reduced graphene oxide surface through oxygen iota of C = O gather and nitrogen molecule of N-H bond coming through urea nano silica suspension. The results found in consistent with the findings of [14 and 42].

3.1.2 X-ray diffractometer (XRD)

XRD pattern of synthesized mNs and rGO composites with urea is shown in Fig. 2a and 2b. Absence of peaks indicative that nano silica synthesized by sol-gel method is amorphous. TGA/DTA curves in Fig. 2.b shows loss of moisture bound water at around 100°C due to phase transformation and loss of crystal bound water. EDX images prove formation of urea-mNS composites with surface elemental composition dominated by Si, C, N and O and urea-rGO composites with surface domination of N, C and O in percent weight and percent atom. Similarly, images of scanning electron microscopy and EDX confirm surface morphology of ordinary mesoporous nano silica (mNs) and urea embedded mNs. This comes out due to vacuum filtration handle and solidify freeze drying handle, which likely causes a halfway restacking of sheets of graphene oxide within fabrication. All through over investigation, number of planes diminished

from mesoporous nano silica and graphene oxide and d-values were found diminish. This might be due to overlapping of planes of urea molecules and thus there is reorientation within structure and brings distinctive morphology of precious material itself. This is very much striking phenomena providing strong evidence of formation of wanted nano composites. These results are in concomitance with findings of [28 ; 19].

3.1.3 Energy Dispersive X-Ray (EDX) Analysis

The X - ray technique, Energy Dispersive X-Ray Analysis (EDX) also known as EDS or EDAX, had been used to determine the elemental composition of nano composites. The study of EDX spectra is given in Fig. 3a and 3b. EDX analysis of mNs samples reveal presence of silicon as well as targeted nutrient including urea molecules. The EDX spectrum of urea-mNs nanocomposites showed 40.24 wt % silicon and 45.10 wt % of Nitrogen content. Urea doped reduced graphene oxide nanocomposites were investigated using EDX spectra, which reveals the presence of Nitrogen (47.54 wt %) and Carbon (37.12 wt %). All of these samples were then processed as a whole and evaluated for elemental composition using manual assessment methods of estimation of nitrogen from synthesized composites as nano fertilizer.

3.1.4 Field Emission Scanning Electron Microscopy (FE–SEM)

FE–SEM was performed using CARL ZEISS SUPRA 55 operating at 10 kv. SEM analysis revealed the surface morphology of spherical nanoparticles of silica in Fig. 4a. Nano silica particles were porous providing the average space of 45–70nm. The SEM micrograph revealed that the addition of urea molecule significantly altered the morphology of silica and graphene nanomaterials. The SEM picture of mesoporous nano silica alone is shown in Fig. 4.a indicating the semi-spheroidal structure of silica nano composites provide spaces for attachment and entrapment of targeted element or material in question. SEM analysis of data (Fig. 4b), indicated that urea- silica complex was formed in spherical shaped and there was aggregation which was predicted as the result of coagulation of small particles after sonication [23]. The particle size depends upon the velocity of stirring process. In the present study the velocity used at 1200. Figure 4.c reveals the lattice fringed structure of reduced graphene oxide with average interlayer spacing of 5–10 nm. In Fig. 5d, the attachment of urea nano particles (might be in the form of inorganic nitrogen species) clearly can be seen. The folded and wrinkled structure provide greater compatibility which makes graphene super loader for the targeted nutrient delivery. The average grain size of 20–70 nm were observed under *Image J* image processor. The results are found in consistent with the results of [28 ; 19].

3.1.5 TEM (Transmission electron microscopy)

The nanocomposites were analyzed with HR-TEM (JEOL JEM 2100 PLUS) with 200 keV acceleration voltage The TEM analysis and respective results are shown in Fig. 5a and b. The silica grains are spherical to cubical in form with average homogeneous particle size distribution of around 50–80 nm, whereas the addition of urea molecules to silica resulted in a small amount of aggregate spherical like

particles with a diameter of about 80 nm that are well bound to the surface of silica nanoparticles. Figure 5c HRTEM micrograph of rGO sheets that clearly displays the graphene lattice fringes. This provides additional detail regarding the rGO material's interplanar distance, which is 5–7 nm. Figure 5d have shown typical TEM micrographs of reduced graphene oxide and urea material recorded for different zones. No graphite-like regions are recorded in any TEM micrographs. Even though according to the results of TEM examination, the graphene based material is nano structured and nano organized at the molecular level with particles of 42–78 nm in size, their aggregates don't possess any translation symmetry characteristic of graphene layers. The results clearly indicate successful synthesis of both the nanocomposites with entrapped nitrogen in composites are achieved through urea seed solution. The results are in agreement with the findings of [57; 27].

3.1.6 SAED imaging

The crystallinity of particles was determined by X-ray diffraction (XRD) "BRUCKER D8 ADVANCE". The crystallinity of both nanocomposites revealed with the help of SAED imaging (Fig. 6a and b). The micrograph in Fig. 6a confirms the polycrystalline to amorphous nature of nano silica based nanocomposite. Similarly micrograph in Fig. 6.b shows crystalline nature of graphene oxide based nanocomposites. The crystal structure and morphology of nano SiO_2 – Urea were studied by XRD. From the XRD pattern (Fig. 2a and b) the characteristic diffraction broad peak centered on 23° (2θ) has been confirmed its amorphous nature. The morphological chart appeared in Fig. 6.a shows that nano SiO_2 is round shape, but did not scatter well. Furthermore, it can be also clearly seen that the average particle size of the nano SiO_2 is approximately 15–20 nm. The precious stone structure and morphology of rGO – Urea were explored by XRD. The peak at $2\theta = 23^\circ$ was as of now anticipated for the GO, due to the increment within inter planar remove between the nanoplatelets after the inclusion of the urea particle (Nitrogen species), as well as the peak at $2\theta = 26^\circ$ with no escalated, alluding to a few remaining graphite platelets. The sequential crests at 44 and 32 credited to the nearness of polycrystalline inorganic nitrogen species cluster of atoms in both of composites (Fig. 6a and b).

3.2 Urea nitrogen loading efficiency of nano composites

To evaluate loading efficacy of urea nitrogen on mesoporous nano silica and reduced graphene oxide, total 12 ratios between nano materials and urea were made (Fig. 1). Positive gradual increase in loading efficiency was observed up to engrafting ratio no. 1 to 8. in which 1: 0.1 to 1:1 (w/w) ratios of mNs / rGO : urea were taken. With increment in targeted compound to be loaded, maximum loading efficacy (58.91% in mNs and 62.73% in rGO) was noted with ratio no. 9 which have 1:1.1 (w/w) ratio of mNs / rGO and urea. The constant loading efficiency was recorded from ratio no. 9 to 12 with average values of 58.91% and 62.73% in case of mNs and rGO, respectively. The constant loading of urea nitrogen in silica minerals might be due to saturation of pore spaces by N in mNs or rGO nanomaterials. These findings are in accordance with the results of [60].

3.3 Dilution effect on nitrogen release by nano composites

To know release pattern of nitrogen from synthesized nano composites of mesoporous silica and reduced graphene oxides, various dilutions (1.10, 1.15, 1.20, 1.40 and 1.50 times) with distilled water were made. Maximum nitrogen content (59.3% in mNs and 62.77% in rGO) were found with 1.5 dilution. Not much remarkable effect on nitrogen release in water by nano composites in water was found with increasing dilutions. Almost constant nitrogen content was recorded with higher dilutions compared to lower dilutions with average value of 58.9 and 62.73% N content in water through synthesized silica and graphene composites (Table 2). This might be ascribed due to controlled release of nitrogen properties of mesoporous silica and reduced graphene as reported by [60].

Table 2
Effects of dilution with distilled water on nitrogen content of nanocomposite

Sl.No	Name of fertilizer	Dilutions(times) with DW	Nitrogen %	Avg. nitrogen %
1	Urea-mNs	1:10	58.80	58.9
2	Urea-mNs	1:15	58.90	
3	Urea-mNs	1:20	58.70	
4	Urea-mNs	1:40	58.80	
5	Urea-mNs	1:50	59.30	
6	Urea-rGO	1:10	62.73	62.73
7	Urea-rGO	1:15	62.70	
8	Urea-rGO	1:20	62.71	
9	Urea-rGO	1:40	62.74	
10	Urea-rGO	1:50	62.77	

3.4 Evaluation of nanocomposite effects on seed germination.

3.4.1 Seed germination and seed vigor index

Figure 8 & 9 have shown the seed germination and seed vigor under various treatments of urea-mNs and urea-rGO. In wheat, all treatments of urea-mNs and urea-rGO had significantly higher seed germination and seed vigor than the control. Treatment T9, had .63 and 213.92 percent more seed germination and seed vigor than the control, respectively [19]. However, treatments T12 and T10 were observed statistically at par with respect to seed germination and seed vigor among themselves. Furthermore, for seed germination and seed vigor, treatment T10 and T9 where nitrogen was applied at 120 and 90 kg N

ha^{-1} through Urea-mNs were significantly higher than T5 and T4 where nitrogen was supplied @ 120 and 90 kg N ha^{-1} through ordinary urea and T13 and T12 where nitrogen was supplied @ 120 and 90 kg N ha^{-1} through urea-rGO composites. The results have also been reported by [8]. Interestingly, germination percentage and seed vigor of rice increased remarkably as the exposure to higher amount of nano composites of mNs. The data pertaining to germination percentage as affected by the different treatments with treated seeds are shown in Figs. 8 and 9. The highest germination and seed vigour index was with [8]. These effects have been attributed to urea itself, to biuret and other impurities in urea fertilizers, and to products formed by hydrolysis or other transformations due entirely to the ammonia produced through the hydrolysis ($\text{NH}_2\text{CO NH}_2 + \text{H}_2\text{O} \rightarrow 2\text{NH}_3 + \text{CO}_2$) of urea. This suggested that these adverse effects could be eliminated or reduced by amending urea fertilizers with small amounts of mNs. A similar study reported that an increase in germination may be due to the absorption and utilization of mNs by seeds [52 ; 53]. According to [52], nano- SiO_2 treatment increased seed germination by providing better nutrients availability to maize seeds and better pH and conductivity to the growing medium.

Our results are in consistent with those of [2] findings, they found that the use of Si and nano SiO_2 significantly improved seed germination characteristics. [7] found that @ low level graphene prevents H_2O_2 over production in faba bean seedlings and as a result low level H_2O_2 accumulated in faba bean may enhance liquid absorption by seeds, improving root emergence and growth of seedling. Rice seed germination, seminal root growth and bud growth all decreased at higher GO concentrations [62].

3.4.2 Impacts of nano composites on rice and wheat crops

3.4.2.1 Grain and straw yields of crops

The highest grain yields over all the treatments in rice and wheat was recorded with the treatment, T11 (120 Kg N ha^{-1} through mNs) followed by T10 and T9, and grain yield of rice under T11 was 64.44, 55.79 and 48% more and 51.19, 34.25 and 32.29% more in wheat compared to T5, T6 and T7 (conventional fertilizer), respectively. The grain yields under the treatment T9 was 38 and 31.58% more in rice and 33.33 and 18.39% more in wheat, respectively comparing the grain yields under T5 and T6, respectively (Fig. 10). The straw yield of rice and wheat was maximum with 120 Kg N ha^{-1} through mNs composites treatment (T11) and resulted in 41.0 and 4.67% increment in rice straw and 128.57 and 76.62% increase in wheat straw over absolute control (T2) and conventional fertilizer (T7), respectively. The grain and straw yield of both crops were statistically at par to each other under T9, T10, T11, T12 and T13 treatments. Compared with treatment T7 (120 kg N ha^{-1} through ordinary urea), T9, T10 and T11 treatments produced significantly higher grain and straw yields of both crops when nitrogen was applied through urea-mNs composites to supply N @ 60, 90 and 120 kg N ha^{-1} , respectively. The increase in crop yields might be ascribed due to nano fertilisers regulate the release of nutrients as needed by crop roots and act as a growth stimulator and carrier for nutrients absorption by plants. These findings are in consistence with [8, 24; 59] who reported that nano fertilizer enhances growth and productivity of wheat plants. Significant increase discovered in rice grain and straw yields with higher silica levels [37, 47; 63].

On contrary to these, higher doses of nitrogen applied through urea-rGO composites @ 120 (T15) and 90 (T14) kg N ha⁻¹ significantly declined the grain yields in both crops compared to all treatments (T8, T9, T10 and T11) and conventional use of urea treatments (T4, T5, T6 and T7). Significantly higher grain yield was recorded with applied @ Nitrogen 30 (T12) and 60 (T13) kg N ha⁻¹ by urea- rGO composites compared with conventional use of urea @ 120 kg N ha⁻¹ (T7). Concerning graphene, the controlled release of nitrogen from rGO increases grain and straw yields at lower doses of nitrogen-rGO and decreasing at higher concentrations [66, 67; 68]. This is due to more exposure to rGO at higher concentration (2000 ppm) resulted in severe losses of crop morphology and a decrease in root and shoots length and biomass, as well as damage to crop tissue structures as reported by [64; 58]. Compared with nano silica (mNs), lower quantity of rGO with urea showed almost at par growth and yields of both crops. Regarding GO, [31] found that graphene oxide could act as a respiratory electron acceptor and promote bacterial culture growth [33] claimed that graphene oxide increased soil bacterial diversity and improved community composition at lower concentration.

3.8.2 Nitrogen content and uptake

Nitrogen content and uptake in table 4 indicated that all treatments of urea-mNs and urea rGO composites had significantly higher N content and total N uptake in rice and wheat compared to control. Treatment T9, T10 and T11 gave 13.85, 15.38 and 22.30% more N content in rice grain and 11.11, 14.82 and 20.74, % more N content in wheat grain, respectively compared to conventional use of urea @ 120kg ha⁻¹ (T7). However, treatments T6, T7, T11 and T12 were at par to each other but significantly lower than T8, T9 and T10 treatments [40]. The increase in N content might be ascribed due to slow and controlled release of nitrogen in soil from engrafted urea- mNs which further increases nutrients absorption by plants. Our results are in agreement with the findings of [30; 51]. Furthermore, nitrogen content with T14 and T15 was significantly lower than all urea-mNs composite treatments (T8, T9 and T10) and conventional urea treatments (T4, T5, T6 and T7). Nitrogen increase with lower doses of urea-rGO treatments is due to higher absorption properties of graphene, whereas higher dose of graphene adversely affects biochemical conversion of nutrients and microbial activity, resulting in decrease in available nitrogen in soil. [6; 44] reported similar results, stating that when graphene reaches up to a certain concentration level in soil (2000 ppm), have toxic effects on N cycle in soil.

Treatment T11 had highest total N uptake (84.52 and 89.80 mg pot⁻¹ by rice and wheat, respectively) followed by T10 and T9 and lowest was in Illrd control (T3) which received maximum amount of rGO. Statistically at par total N uptake was with T9, T12 and T13 treatments to each other and significantly higher than T3, T4, T5, T14 and T15 treatments. 17.08, 47.70 and 59.77% more total N uptake by rice and 47.94, 62.38 and 88.61% more total N uptake by wheat was exhibited with T9, T10 and T11 treatments, respectively compared to T6 treatment (120 kg N ha⁻¹ through ordinary urea). Similarly, comparing to T6 Treatment, T12 and T13 showed 25.31 and 25.99% increase in total N uptake of rice and 51.50 and 53.07% increase in total N uptake of wheat. The increase in total N uptake by crops might be ascribed due to controlled release and increased nitrogen use efficacy at lower doses of graphene, whereas higher doses of mNs are beneficial to nutrients cycling and microbial activities, resulting in higher available

nitrogen in soil. Similar findings to ours have been reported by [34] who found that increasing graphene in soil has toxic impact on microbial activities. The use of nano silica increased the available N, P_2O_5 and K_2O content in soil [30 ; **51**]. However, treatments T9, T10, T11 and T12 gave significantly higher total nitrogen uptake with urea-mNs composites compared to T5 and T4, in which nitrogen was applied @ 120 and 90 kg N ha⁻¹ by ordinary urea; and T13 and T12, in which N was applied at 120 and 90 kg N ha⁻¹ through urea-rGO composites.

Table 3 : Effect of varying levels of N through ordinary Urea, Urea-mNs and Urea-rGO on Nitrogen content and uptake by grain and straw of wheat.

Treatment	N Content (%)				Total N uptake (Grain +Straw) (g pot ⁻¹)	
	Rice		Wheat		Rice	Wheat
	Grain	Straw	Grain	Straw		
T1– N0P0K0+Maxi. mSN	0.92	0.46	1.25	0.50	24.70	16.20
T2– N0P0K0+Maxi. mSN	0.92	0.46	1.25	0.50	22.62	17.53
T3 - N0P0K0 +.Maxi. rGO	0.90	0.35	0.81	0.31	15.60	11.01
T4 - N ₃₀ (urea) P ₆₀ K ₄₀	1.10	0.50	1.14	0.50	32.05	30.45
T5 - N ₆₀ (urea) P ₆₀ K ₄₀	1.20	0.53	1.34	0.53	41.38	37.29
T6 - N ₉₀ (urea) P ₆₀ K ₄₀	1.25	0.55	1.38	0.55	47.47	43.93
T7 - N ₁₂₀ (urea) P ₆₀ K ₄₀	1.30	0.59	1.35	0.59	52.90	47.61
T8 - N ₃₀ (Urea- mNs) P ₆₀ K ₄₀	1.25	0.56	1.49	0.56	42.05	57.34
T9 - N ₆₀ (Urea- mNs) P ₆₀ K ₄₀	1.48	0.60	1.55	0.60	61.94	69.96
T10 - N ₉₀ (Urea- mNs) P ₆₀ K ₄₀	1.50	0.65	1.58	0.65	76.02	77.31
T11 - N ₁₂₀ (Urea- mNs) P ₆₀ K ₄₀	1.59	0.69	1.63	0.69	84.52	89.80
T12- N ₃₀ (Urea-rGO) P ₆₀ K ₄₀	1.38	0.68	1.40	0.68	66.29	72.13
T13- N ₆₀ (Urea-rGO) P ₆₀ K ₄₀	1.34	0.63	1.38	0.63	66.65	72.88
T14- N ₉₀ (Urea-rGO) P ₆₀ K ₄₀	1.10	0.57	1.15	0.57	27.47	39.01
T15- N ₁₂₀ (Urea-rGO) P ₆₀ K ₄₀	1.00	0.55	1.09	0.55	23.34	28.96
SE(m) ±	0.04	0.01	0.02	0.01	1.022	1.631
C.D.	0.19	0.04	0.05	0.04	2.743	3.876

3.8.3 Nitrogen Use efficiency (NUE)

The highest nitrogen use efficiency (75.71% in rice and 91.36% in wheat) urea treatments was with T12 followed by T13 and T8, by giving 38.12 and 29.04% NUE in rice and 46.20 and 59.50% NUE in wheat, respectively. Conventional urea treatments (T4,T5,T6 and T7) recorded much lower NUE compared to

urea mNs and rGO composite treatments except T14 and T5. Lowest NUE was with T14 and T15 treatments which received 90 and 120 kg N ha⁻¹ through rGO. All treatments of graded levels of nano nitrogen showed improved N use efficiency over conventional fertilization (T2). The maximum fertilizer use efficiency (FUE) for Fe was recorded with T3, while the maximum FUE of Zn was found under T4 (Fig. 6).

3.3 Soil Properties

3.3.1 Soil pH, Electrical conductivity and Organic carbon

The data on pH, EC and Organic carbon indicate that urea–mSN and graphene composites do not have remarkable change in pH, electrical conductivity (EC) and organic carbon in soil except slight reduction in soil pH with increasing the amount of Urea-rGO composites (Table 5). However, urea–mNs composites treatments showed slight enhancement in organic carbon content.

3.3.2 Available nitrogen, phosphorus and potassium

After harvest, different treatments of Urea-mNs and urea-rGO have shown a significant impact on available nitrogen, phosphorus and potassium in soil. The initial soil contained 260, 14.2 and 110.7 kg ha⁻¹ nitrogen, phosphorous and potassium which was reduced to 255.0, 13.1 and 108.5.0 in IInd control after wheat cropping which maximum amount of graphene were added. Significantly higher values of nitrogen, phosphorous and potassium in soil after wheat harvest were found with T8, T9, T10, T11 and T12 treatments compared with T6 (120 kg N ha⁻¹ by ordinary urea) including all other silica and graphene composite treatments. The increase in available N, P and K in soil might be ascribed due to excess accumulation of applied nitrogen through Urea-mNs and urea-rGO. These findings are consistent with those of [18] who found that nano silica has a high dissolution rate and a salty nature, which may be attributed to an increase in soil nitrogen. T13 and T14 treatments, in which nitrogen was applied @ 90 and 120 kg N ha⁻¹ through urea-rGO resulted significantly lower available N, P and K in soil compared to T8 and T9 treatments. When urea-mNs was used as N source, all treatments had significant increase in available N, P and K in soil over convetional urea treatmem T6 (@ 120 kg ha⁻¹) but in case of urea-rGO, lower doses of nitrogen (30 and 60 kg N ha⁻¹) gave significantly higher available N, P and K in soil as compared to T13 and T14 treatment that exhibited significantly lower N, P and K compared to T8,T9 and T10 treatments. The reduction in available N, P and K was appears due to toxic effect of rGO on root biomass in soil at higher concentrations as reported by [36]

Table 4 : Effect of varying levels of N through ordinary Urea, Urea-mNs and Urea-rGO on soil properties after wheat harvest

Treatment	Soil properties after rice–wheat cropping					
	pH (1:2.5)	E.C (dSm ⁻¹)	O.C (%)	Available nitrogen (kg ha ⁻¹)	Available phosphorus (kg ha ⁻¹)	Available potassium (kg ha ⁻¹)
T1– N0 P ₆₀ K ₄₀ + Maxi. mSN	8.16	0.24	0.79	252.0	14.2	110.7
T2- N0 P ₆₀ K ₄₀ + Maxi. rGO	8.10	0.22	0.70	250.0	13.1	108.5
T3 – N0 P ₆₀ K ₄₀	8.16	0.44	0.79	257.0	14.8	115.7
T4 - N _{30(urea)} P ₆₀ K ₄₀	8.16	0.44	0.79	257.0	14.8	115.7
T5 - N _{60(urea)} P ₆₀ K ₄₀	8.13	0.44	0.79	267.5	14.9	118.9
T6 - N _{90(urea)} P ₆₀ K ₄₀	8.18	0.45	0.80	268.0	15.0	122.0
T7 - N _{120(urea)} P ₆₀ K ₄₀	8.22	0.45	0.83	279.0	15.5	125.6
T8 - N _{30(Urea– mNs)} P ₆₀ K ₄₀	8.19	0.45	0.81	275.0	14.1	123.7
T9 - N _{60(Urea– mNs)} P ₆₀ K ₄₀	8.15	0.45	0.82	279.6	16.5	133.9
T10 - N _{90(Urea– mNs)} P ₆₀ K ₄₀	8.16	0.45	0.85	283.0	16.9	139.0
T11 - N _{120(Urea– mNs)} P ₆₀ K ₄₀	8.17	0.46	0.86	289.5	17.9	142.4
T12- N _{30(Urea–rGO)} P ₆₀ K ₄₀	8.15	0.46	0.83	280.0	16.0	135.6
T13- N _{60(Urea–rGO)} P ₆₀ K ₄₀	8.10	0.46	0.84	283.5	16.6	138.4
T14- N _{90(Urea–rGO)} P ₆₀ K ₄₀	8.12	0.46	0.81	279.0	15.2	124.6

Treatment	Soil properties after rice–wheat cropping					
	pH (1:2.5)	E.C (dSm ⁻¹)	O.C (%)	Available nitrogen (kg ha ⁻¹)	Available phosphorus (kg ha ⁻¹)	Available potassium (kg ha ⁻¹)
T15- N ₁₂₀ (Urea-rGO) P ₆₀ K ₄₀	8.13	0.47	0.80	277.0	15.0	122.6
SE(m) ±	0.003	0.002	0.001	3.007	0.301	2.042
C.D.	NS	NS	NS	9.302	1.032	7.165
Initial Soil	7.95	0.20	0.78	263.7	16.31	126.4

3.3.3 Biological Properties

3.3.3.1 Soil microbial population and dehydrogenase activity

All treatment combinations of urea-mNs and urea-rGO had significantly higher microbial population and dehydrogenase activity over control (tables 6). Maximum 61.89, 127.17, 89.75 and 123.87 percent increase over the control in bacterial, fungal, actinomycetes population and dehydrogenase activity, respectively. Furthermore, for microbial population and dehydrogenase activity in treatment T7, T8 and T9, where N was applied @ 30, 60 and 90 kg N ha⁻¹ through urea-mNs were significantly higher as compared to T4, T5 and T6 where nitrogen was applied @ 60,90 and 120 kg N ha⁻¹ through ordinary urea. Treatment T13 and T14 where nitrogen was applied @ 90 and 120 kg N ha⁻¹ through urea-rGO resulted significant reduction in microbial population and dehydrogenase activities in comparison to T8, T9 and T10 treatments. Our results are in agreement with the findings of [30, 65 and 37] found that nano silica do not show any toxic effect on soil bacterial community as compared with other silicon materials. Nano silica enhances the growth of PGPR and increases total soil microbial population. Thus, nano silica can be included for fertilizer formulations to make the soil more productive and to improve beneficial soil microbial community for better yield. These results are in agreement with the findings of [13; 59]. Whereas, use of Urea- rGO composit showed slight reduction on microbial population and dehydrogenase activity in soil. [22, 43] found that the composition of soil fungal communities were significantly influenced by the addition of graphene oxide (GO) and graphite. Whereas, at higher concentrations graphene oxide (GO) can suppress bacterial [5] and fungal cultures [10] by inducing oxidative stress [4] or penetrating cell-walls and extracting phospholipids [55].

Table 5 : Effect of varying levels of N through ordinary Urea, Urea-mNs and Urea-rGO on soil microbial population and Dehydrogenase activity.

Treatment	Total bacterial count (X 10 ⁸ cfu g ⁻¹ dry soil)	Total fungal count (X10 ⁴ cfu g ⁻¹ dry soil)	Total actinomycetes count (X10 ⁵ cfu g ⁻¹ dry soil)	DHA (ug TPF 24 h ⁻¹ g ⁻¹ soil)
T1–N0P0K0 + Maximum mSn	6.22	7.47	10.05	111.67
T2-N0P0K0 + Maximumum rGO	4.34	5.44	8.12	95.55
T3 - N ₃₀ (urea) P ₆₀ K ₄₀	7.21	8.41	11.13	125.00
T4 - N ₆₀ (urea) P ₆₀ K ₄₀	7.27	8.50	11.40	133.33
T5 - N ₉₀ (urea) P ₆₀ K ₄₀	7.24	8.66	11.55	142.67
T6 - N ₁₂₀ (urea) P ₆₀ K ₄₀	7.36	9.65	12.30	160.33
T7 - N ₃₀ (Urea– mNs) P ₆₀ K ₄₀	8.19	12.20	13.20	180.67
T8 - N ₆₀ (Urea– mNs) P ₆₀ K ₄₀	8.60	14.20	14.50	199.00
T9 - N ₉₀ (Urea– mNs) P ₆₀ K ₄₀	8.96	15.33	15.20	220.00
T10 - N ₁₂₀ (Urea– mNs) P ₆₀ K ₄₀	10.07	16.97	19.07	250.00
T11- N ₃₀ (Urea–rGO) P ₆₀ K ₄₀	8.96	14.23	16.20	231.33
T12- N ₆₀ (Urea–rGO) P ₆₀ K ₄₀	9.83	15.47	18.27	200.00

Treatment	Total bacterial count (X 10 ⁸ cfu g ⁻¹ dry soil)	Total fungal count (X10 ⁴ cfu g ⁻¹ dry soil)	Total actinomycetes count (X10 ⁵ cfu g ⁻¹ dry soil)	DHA (ug TPF 24 h ⁻¹ g ⁻¹ soil)
T13- N ₉₀ (Urea-rGO) P ₆₀ K ₄₀	8.05	13.70	15.20	195.33
T14- N ₁₂₀ (Urea-rGO) P ₆₀ K ₄₀	7.52	11.77	14.27	181.67
SE(m) ±	0.09	0.15	0.14	3.05
C.D.	0.26	0.44	0.41	8.90

4. Conclusions

FTIR analysis of mNs-urea and rGO-urea composites indicate nearness of nitrogen (urea) particle by wide peaks at 3348 cm⁻¹ and 3334 cm⁻¹ in tall recurrence region due to apparent crest of N-H starching vibrations. The SEM micrograph reveals that the addition of urea molecule significantly altered the morphology of silica and graphene nanomaterials. Compared to control, synthesized composites of mesoporous nano silica and graphene with urea enhance all growth and yield parameters of rice and wheat both crops. Therefore, **nano nitrogen composite capsules** can be used to supply nitrogen as fertilizer to enhance crop growth and yields. Maximum increase in crop yields over conventional urea (@ 120 kg N ha⁻¹) was with urea-mNS composite capsules which were applied to supply 120 kg N ha⁻¹ followed by 90 and 60 kg N ha⁻¹. Urea-mNs and Urea-rGO composites application for 60 kg N ha⁻¹ give significantly higher and satisfactory yield levels of both crops compared to conventional use of ordinary urea (120 kg N ha⁻¹). Lower doses (30 and 60 kg N ha⁻¹) through urea-rGO composites shows significant increase in crop yields over control but higher doses (90 and 120 kg N per hectare) declines crop yields remarkably. Thus, 50% nitrogen and one fourth bulk volume of urea can be saved by this fabricating nano technology that will certainly decline remarkable cost of crop production.

Abbreviations

mNs

Mesoporous nano silica

rGO

reduced graphene oxide

FTIR

These composites were characterized by using Fourier Transform Infrared Spectroscopy
XRD
X-ray diffractometer
SEM
Scanning electron microscopy
TEM
Transmission electron microscopy
EDX
Energy-dispersive X-ray

Declarations

Data Availability

All data are presented within the article.

Competing Interests

The authors declare that they have no conflicts of interest.

Author's Contribution

Ram S and Mubasir S. carried out the synthesis of nano composites and Shubham A D and Singh K were involved in acquisition of data, analysis and interpretation to draft manuscript. The former authors gave final approval of the version to be published. All authors read and approved the final manuscript

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Figures

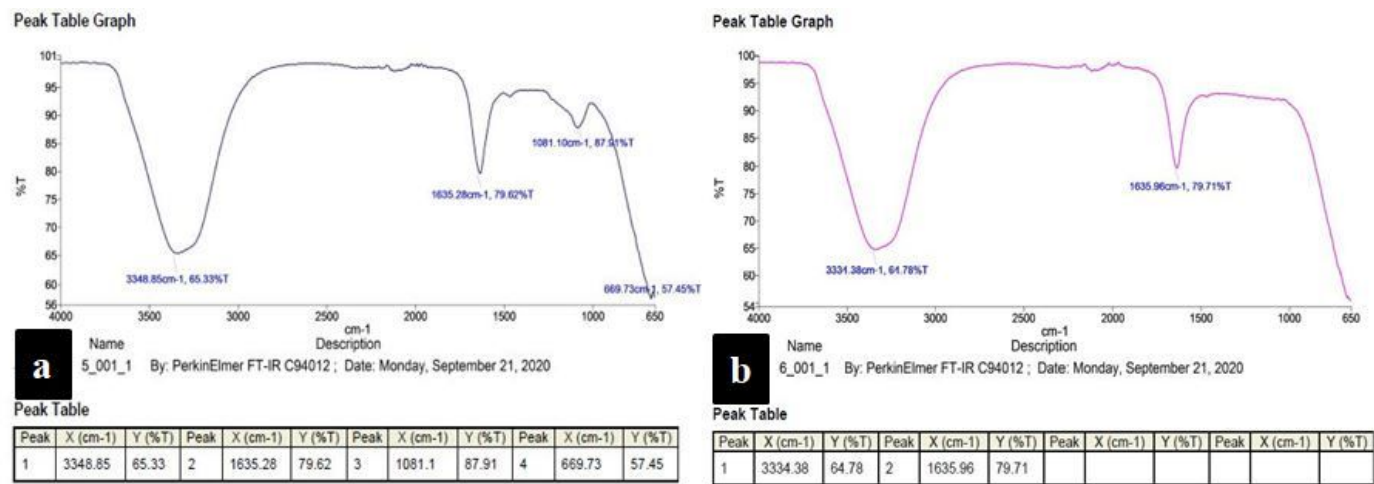


Figure 1

FTIR a. mNS-Urea nanocomposite and b. rGO-Urea nanocomposite

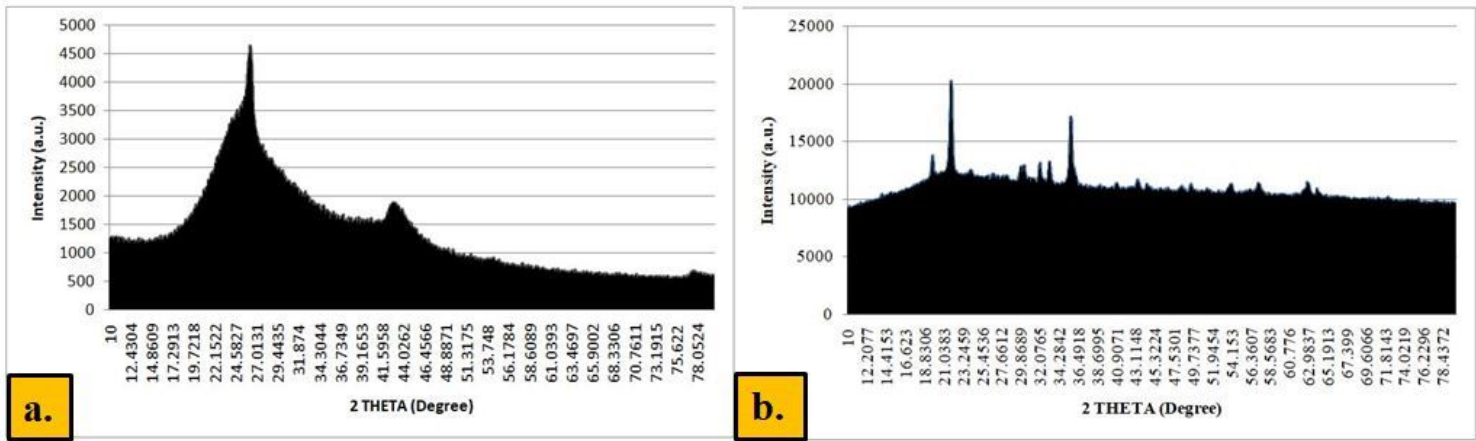


Figure 2

XRD spectra of a. mNS-Urea nanocomposite and b. rGO-Urea nanocomposite

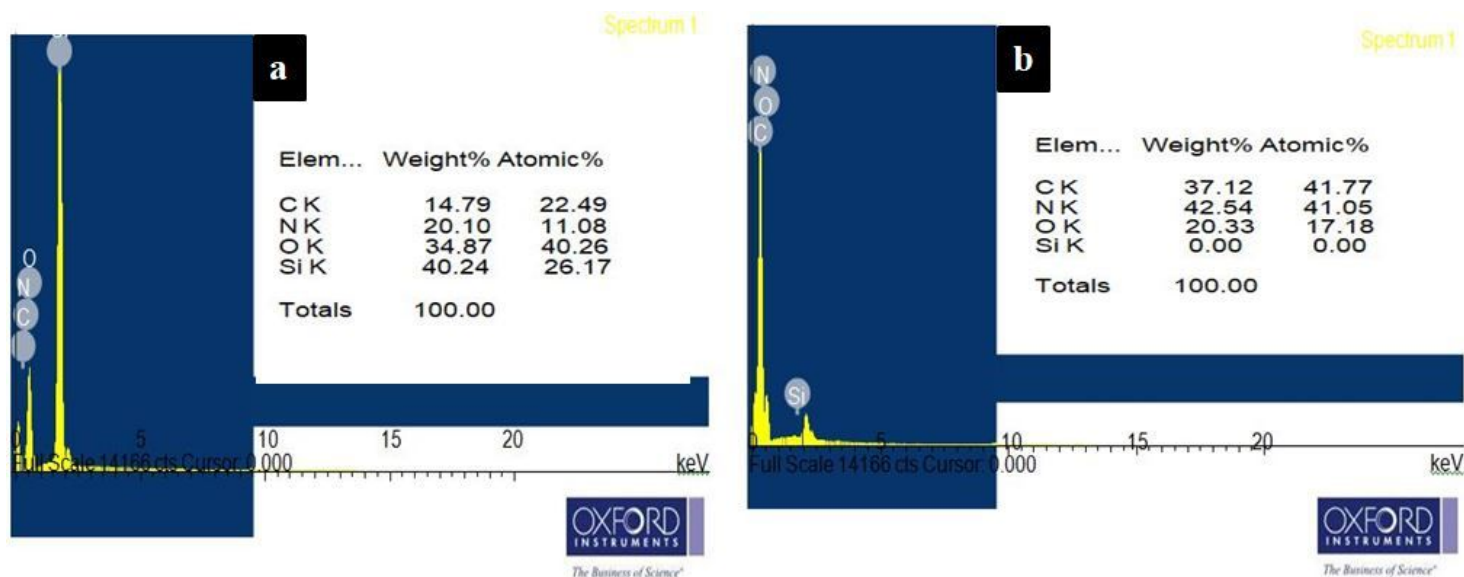


Figure 3

EDX spectra of a. mNs-Urea nanocomposite and b. rGO-Urea nanocomposite

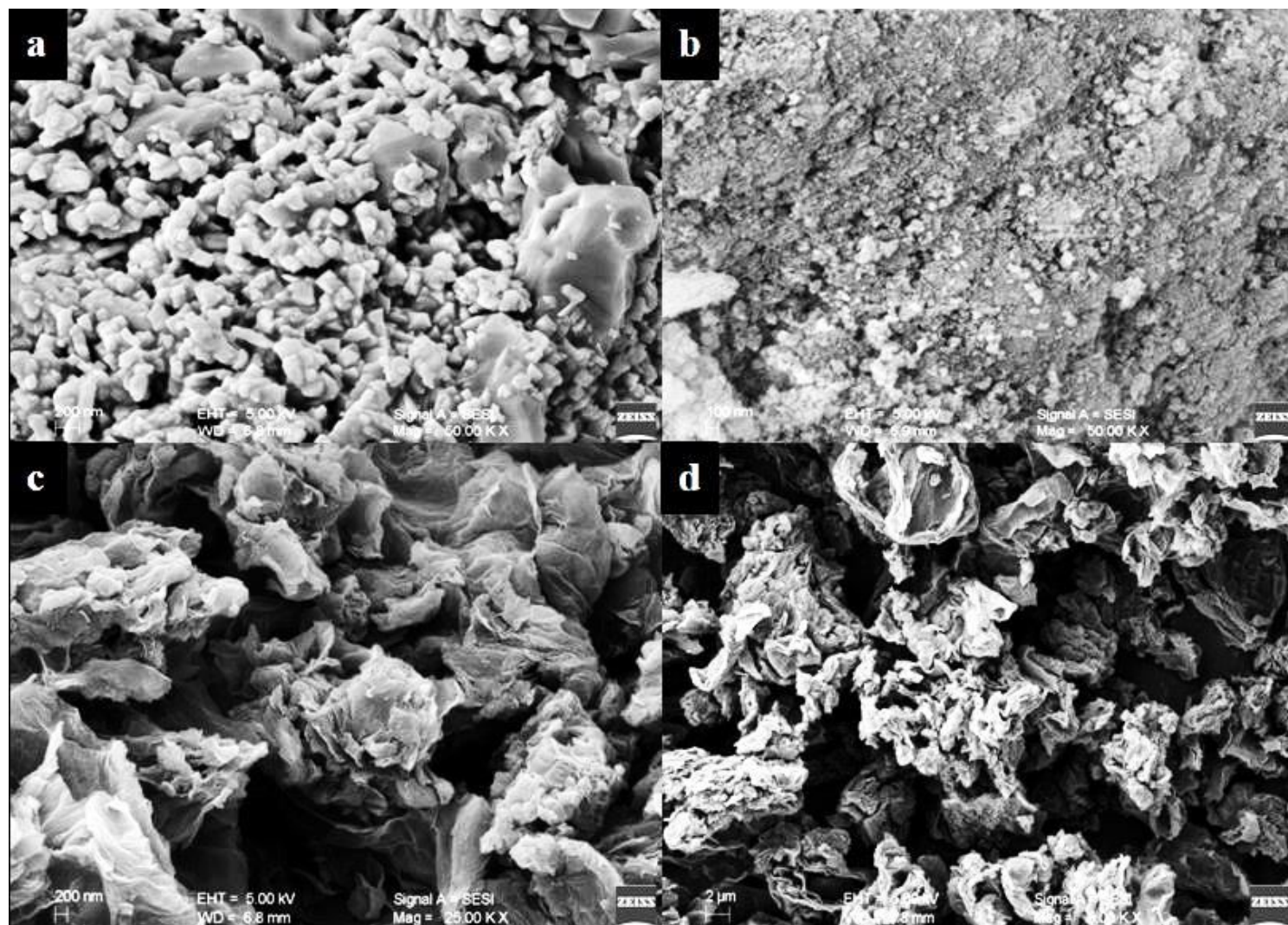


Figure 4

SEM micrograph of a. mNs, b. mNs-Urea nanocomposite, c. rGO and d. rGO-Urea nanocomposite

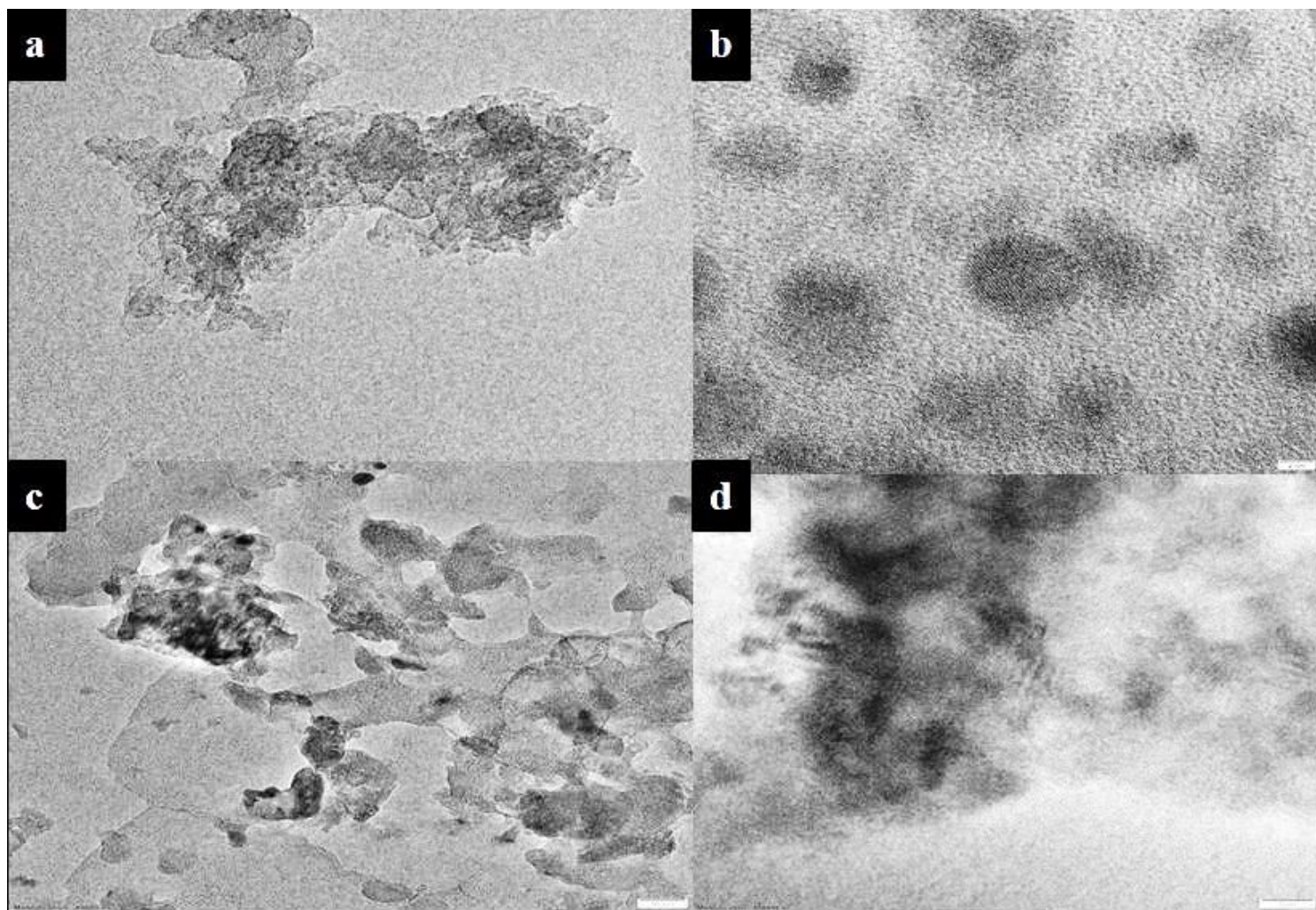


Figure 5

TEM micrograph of a. mNs alone b. mNs-Urea nanocomposite c. rGO alone and d. rGO-Urea nanocomposite

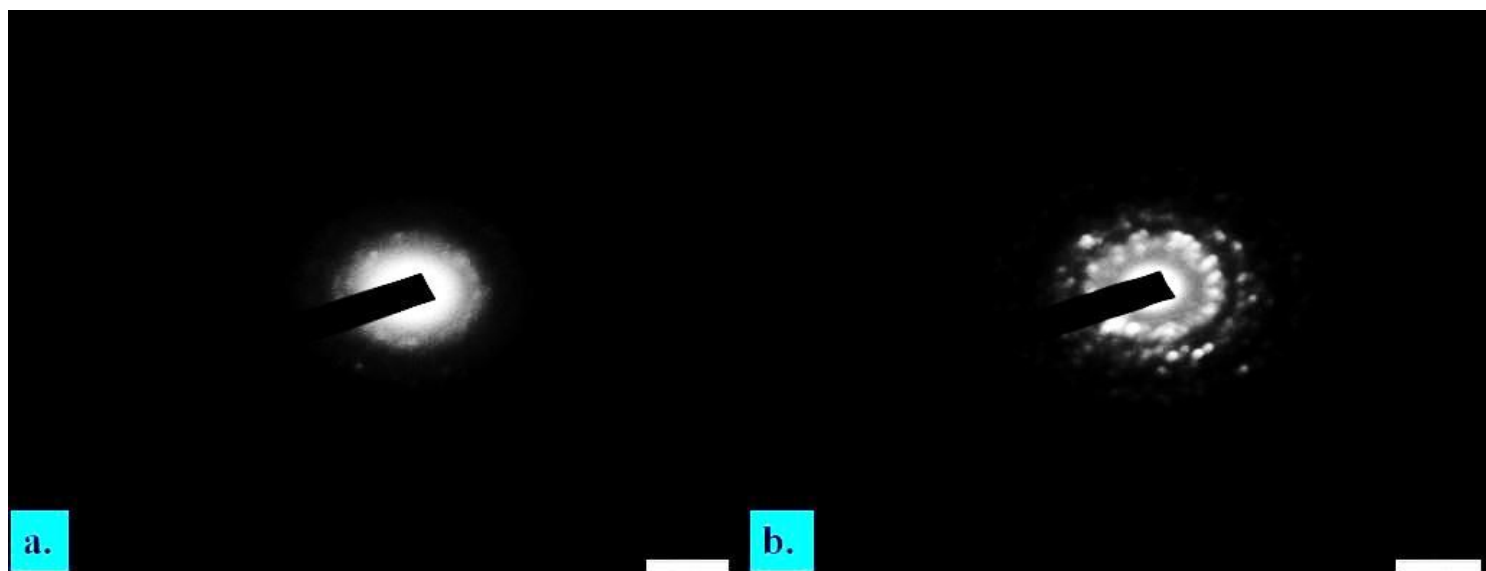


Figure 6

SAED micrograph of a. mNs-Urea nanocomposite and b. rGO-Urea nanocomposite

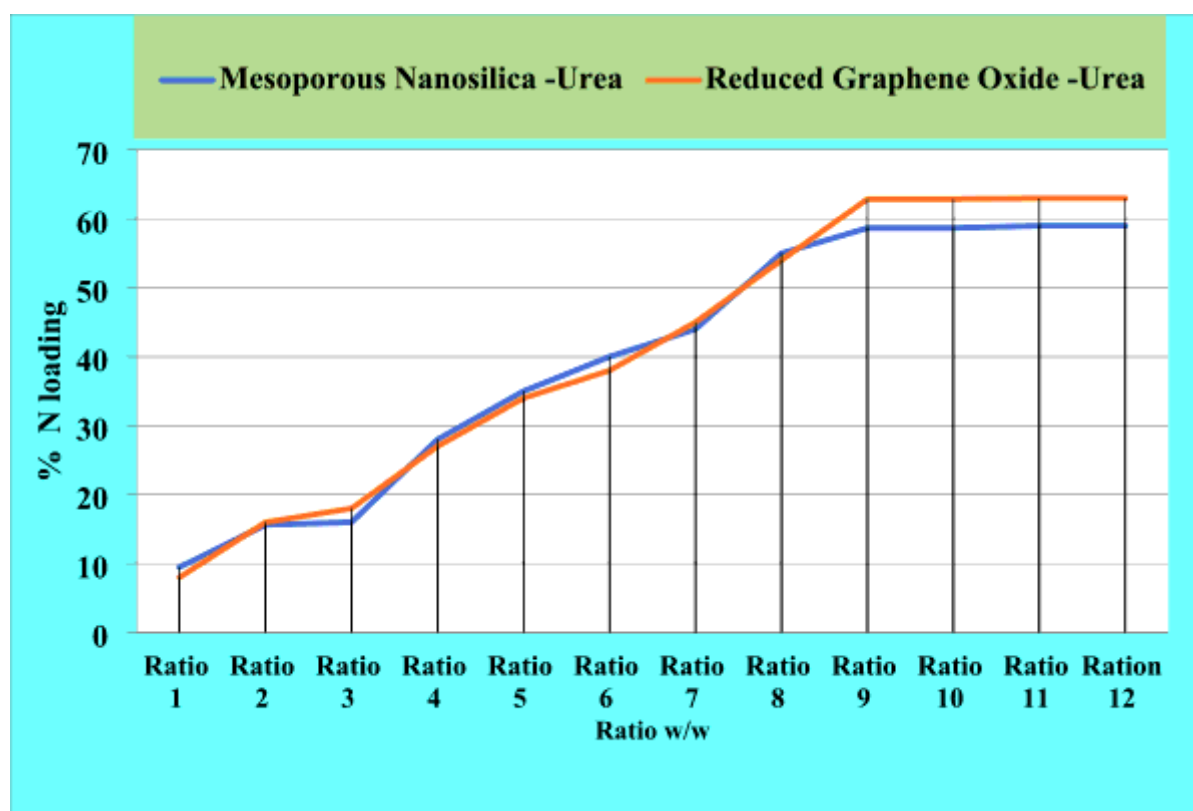


Figure 7

Loading efficacy graphical representation.

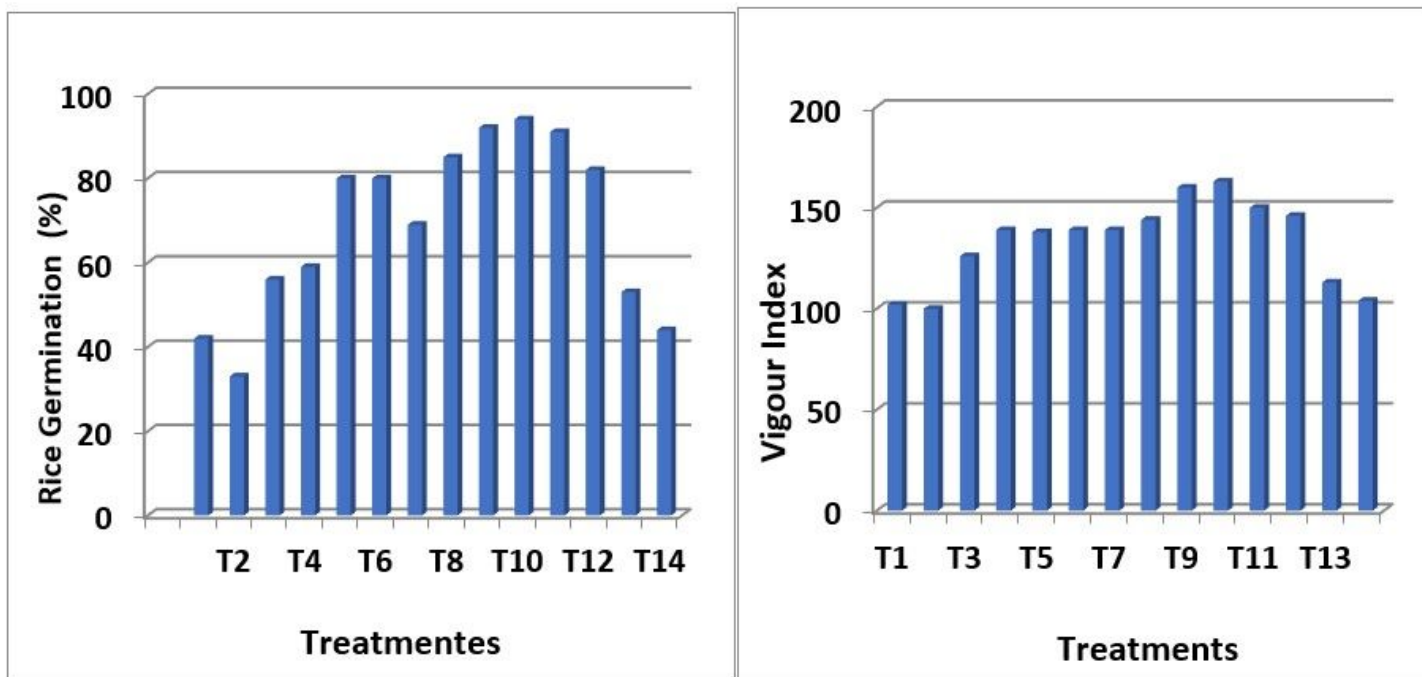


Figure 8

Effect of nano composites on of Silica and rGO composites on rice

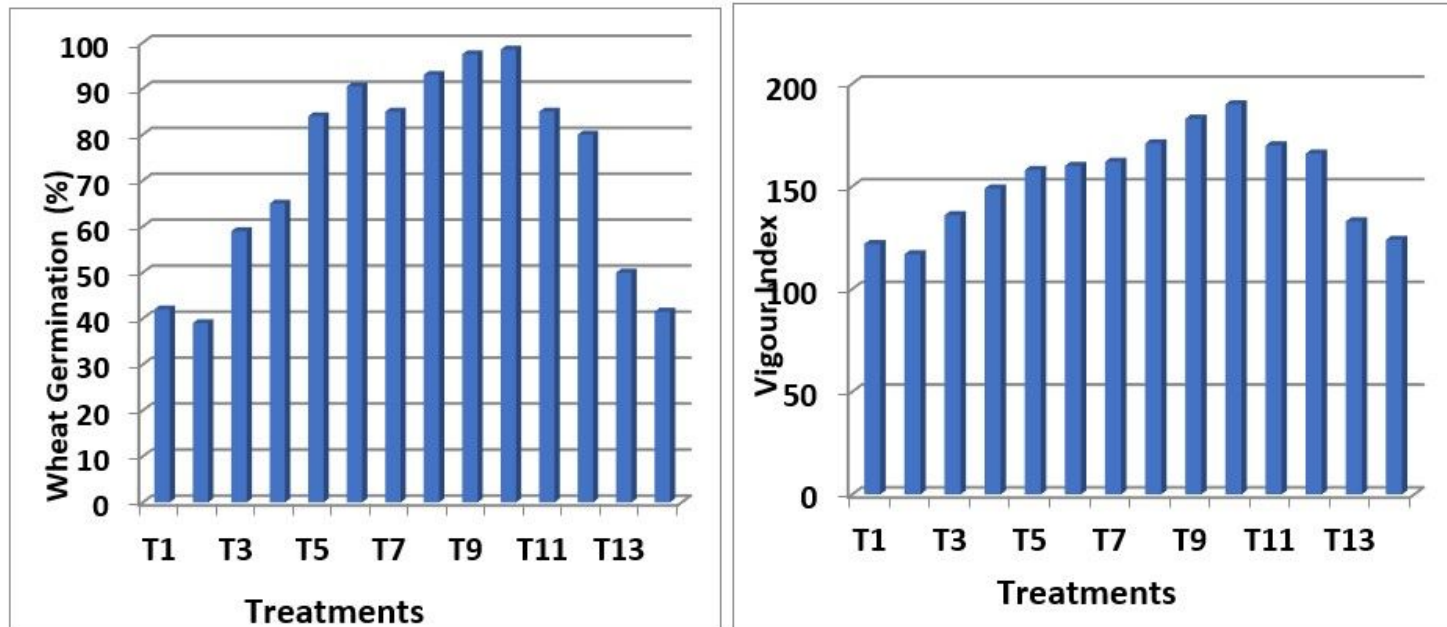


Figure 9

Effect of nano composites on of Silica and rGO composites on wheat

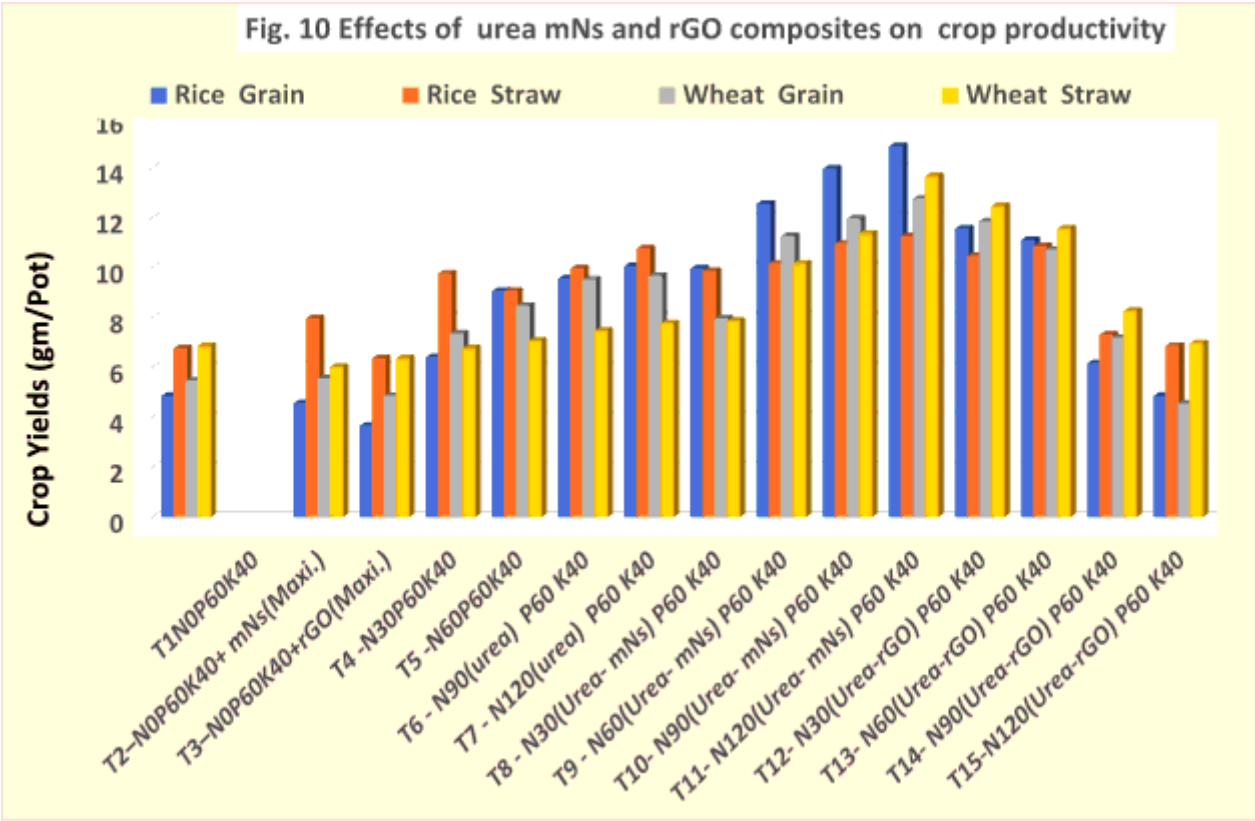


Figure 10

Effect of varying levels of N through ordinary Urea, Urea-mNs and Urea-rGO on grain and straw yields (g/pot).

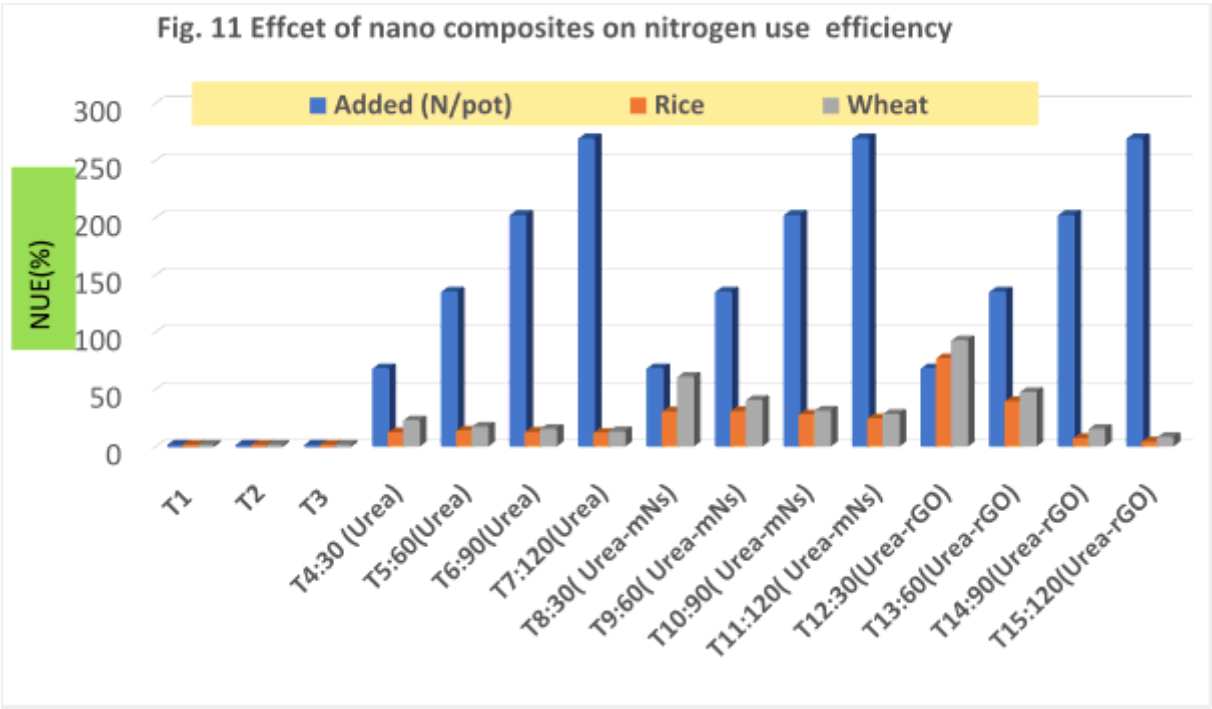


Figure 11

Effect of ordinary Urea, Urea-mNs and Urea-rGO on NUE in rice and wheat crops.